

Explaining Neural Style Transfer

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Abstract—This extended abstract serves as an entry point into our immersive article, allowing readers to explore the concept of neural style transfer in various interactive components. Throughout the article, the task of style transfer and how it can be accomplished by using deep neural networks are introduced. Furthermore, several extensions of the style transfer process such as color correction models are highlighted and described in more detail in the full immersive article

I. INTRODUCTION

Style transfer is a technique that allows users to extract the style of a source image and transfer it to a target image. While the content of the target image should be preserved, its style is preferred to match the style of the source image. An example of this process is shown in Figure 1. This article begins with a closer look at the definition of style transfer and how it differs from texture transfer in Section II. In Section III, style transfer is defined in terms of several loss functions, which are used to train a deep neural network. Furthermore, model optimization-based methods are introduced in Section IV. Finally, an example is provided to show how this approach can be extended to gain more control over the output in Section V.

II. CONCEPT OF NEURAL STYLE TRANSFER

In style transfer, the focus is on transferring high-level visual features such as the color palette, brush strokes, and overall aesthetic of the source image to the target image while ensuring the integrity of the content of the target image. Before the advancement of deep learning, an extensive range of non-parametric algorithms for style modification and transfer has been developed. In contrast, neural style transfer utilizes deep neural networks, typically convolutional neural networks (CNNs) [1], to achieve the desired result. CNNs are composed of multiple layers, each of which is responsible for a specific task. Due to the flexibility of CNNs, Gatys et al. [2] proposed the first CNN-based style transfer method in 2016.

The workflow of neural style transfer is shown in Figure 1. Given a content image c and a style image s , our goal is to generate a stylized image g , which is initialized by random noise. The content features are extracted from c and compared

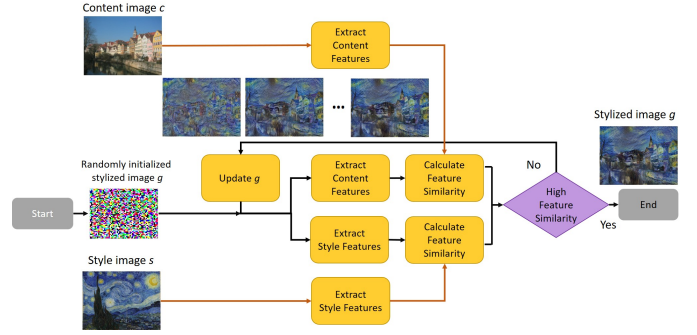


Fig. 1. The workflow of the neural style transfer.

with the content features extracted from c to maximize the content feature similarity. Similarly, the style features are also extracted from g and compared with the style features extracted from s to maximize the style feature similarity. g is then iteratively updated, as shown in Figure 1, to obtain the neural style transfer results. The training process terminates once the extracted content features and style features of g are similar enough to those of c and s .

III. NEURAL NETWORK-BASED STYLE TRANSFER

To generate an image that blends the features of both the style image and content image (source image and target image, respectively), the similarity of their image features needs to be measured. Overall, the generated image should have minimal content-wise differences with respect to the content image and should have similar features with respect to the style image. This is accomplished by defining a total loss function L_{total} that incorporates the weighting parameters (α and β) between the content loss $L_{content}$ of g and c , and the style loss L_{style} of g and s as follows:

$$L_{total}(g, c, s) = \alpha L_{content}(g, c) + \beta L_{style}(g, s). \quad (1)$$

For simplicity, the loss functions are denoted only by their function names throughout the remaining article.

A. Content Preservation

First, the content representation is emphasized. The content of the target image (the image to which the style is applied)

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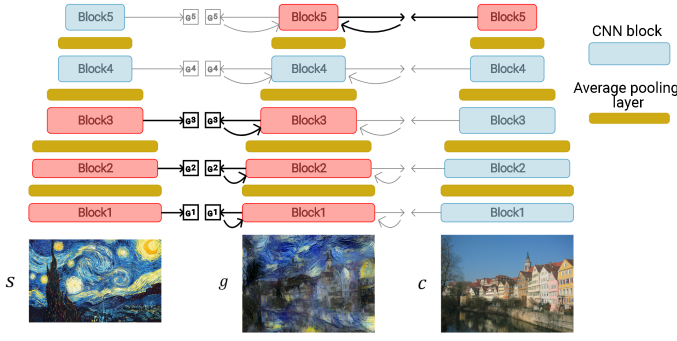


Fig. 2. Image optimization-based style transfer. The CNN-based model allows to select the layer(s) to which the transfer is applied.

should be preserved. Given the content image c , our goal is to generate an image g that has similar content to c . Similar images would mean similar activations in the feature maps of g and c extracted by two CNNs. One CNN receives the content image c , and the other CNN starts with a white noise image g . The white noise image is updated to minimize $L_{content}$, which is calculated based on the squared difference between the feature maps of c and g as:

$$L_{content} = \frac{1}{2} \sum_{ij} (A_{ij}^l(g) - A_{ij}^l(c))^2, \quad (2)$$

where $A_{ij}^l(g)$ and $A_{ij}^l(c)$ are the activations of the i^{th} filter at position j in layer l of g and c , respectively. By minimizing this loss, the generated image g will gradually match the feature representation of the original content image c .

B. Style Analysis

Measuring the difference in content has been proven easy because it can be based on the actual difference of activations in the feature maps. Measuring an image's style is more intricate. Here, the correlations of different feature maps are taken into account and compared among the style image s and the generated image g . This is done by calculating the Gram matrix of the feature maps, which is the inner product between feature maps in the CNN and is defined as:

$$G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l, \quad (3)$$

where G_{ij}^l is the Gram matrix of layer l , F_{ik}^l is the activation of the i^{th} filter at position k in layer l of the processed image. Furthermore, the differences of the Gram matrices of the generated image g and the style image s over all layers and filters are averaged to define the style loss as follows:

$$L_{style} = \frac{1}{(2N^l M^l)^2} \sum_l w_l \sum_{i,j} (G_{ij}^l(g) - G_{ij}^l(s))^2, \quad (4)$$

where $G_{ij}^l(s)$ and $G_{ij}^l(g)$ are the Gram matrices of the style image and the generated image, respectively; w_l is the weight of layer l , N^l is the number of feature maps in layer l , and M^l is the height times the width of the feature map.

C. Combining Content and Style

Finally, the content and style losses are combined to define the total loss as shown in (1). Figure 2 shows the combination of the content and style losses at different levels of the involved CNNs in the deep neural network-based style transfer model.

IV. MODEL OPTIMIZATION-BASED METHOD

Image optimization-based style transfer comes with high computational costs. In contrast, model optimization-based methods such as [3] efficiently generate the stylized image by using a single forward pass of trained networks based on content and style images. A decoder network is trained at first to invert different levels of the encoder features of the encoder network to the original images. After training, the parameters of the encoder and decoder networks are fixed. During style transfer, both content and style images are provided as input to the encoder network to extract content and style features.

V. EXTENSIONS OF STYLE TRANSFER

In order to better represent certain aspects of the input image or to target a more specific style in the generated image, the general process of style transfer can be extended. These extensions allow users to have more control over the output image and the style transfer process. In this section, the concept of color correction as an exemplary extension is briefly presented, while the immersive article provides more information on spatial control and shows how to implement style transfer for photorealistic images.

Color (style) and content can often be considered as independent components. In the previous example, the style transfer process may copy the color distribution of the style image to the content image directly (cf. Figure 2). In some cases, this color distribution may not be desirable for the content. Therefore, our goal is to further separate these two components and provide users with more control over the generated image. Reasonable results have been achieved by either matching the color histograms before the style transfer process or using a luminance-only transfer [4].

VI. CONCLUSION

In this article, deep neural network-based methods to perform style transfer are reviewed. Based on the example of color correction, users can achieve more control over the stylization process. As the length of this extended abstract is limited, the readers are encouraged to study the full immersive article, which provides more details on the style transfer process and its extensions.

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